

## Combined 8-PPT Forward and Inverse Fast Fourier Transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \dots, N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j\frac{2\pi}{N}kn}, \quad n = 0, 1, \dots, N-1$$

$$\boxed{x[n] = \text{IFFT}(X[k]) = \frac{1}{N} \overline{\text{FFT}(X[k])}}$$